

作业 2 答案

1. 已知特性阻抗为 50Ω 的传输线，终端负载为 $Z_L = 100\Omega$ 。求：(1) 终端的反射系数；(2) 距终端 $l = \lambda/8, \lambda/4, 3\lambda/8, \lambda/2, \lambda$ 处的输入阻抗。

解：(1) 终端反射系数： $\Gamma_{uL} = \frac{100 - 50}{100 + 50} = \frac{1}{3}$

(2) 输入阻抗： $Z_{in} = Z_c \frac{Z_L + jZ_c \tan(kl)}{Z_c + jZ_L \tan(kl)}$

当 $l = \lambda/8$ 时， $Z_{in} = Z_c \frac{Z_L + jZ_c \tan(k \frac{\lambda}{8})}{Z_c + jZ_L \tan(k \frac{\lambda}{8})} = 50 \times \frac{2 + j}{1 + j2} = (40 - j30)\Omega$

当 $l = \lambda/4$ 时， $Z_{in} = Z_c \frac{Z_L + jZ_c \tan(k \frac{\lambda}{4})}{Z_c + jZ_L \tan(k \frac{\lambda}{4})} = \frac{Z_c^2}{Z_L} = 25\Omega$

当 $l = 3\lambda/8$ 时， $Z_{in} = Z_c \frac{Z_L + jZ_c \tan(k \frac{3\lambda}{8})}{Z_c + jZ_L \tan(k \frac{3\lambda}{8})} = 50 \times \frac{2 - j}{1 - j2} = (40 + j30)\Omega$

$l = \lambda/2$ 时， $Z_{in} = Z_c \frac{Z_L + jZ_c \tan(k \frac{\lambda}{2})}{Z_c + jZ_L \tan(k \frac{\lambda}{2})} = Z_L = 100\Omega$

$l = \lambda$ 时， $Z_{in} = Z_c \frac{Z_L + jZ_c \tan(k\lambda)}{Z_c + jZ_L \tan(k\lambda)} = Z_L = 100\Omega$

2. 特性阻抗为 $Z_c = 100\Omega$ 的传输线，终端接负载 $Z_L = 100 + j50\Omega$ ，波长为 $\lambda = 1\text{m}$ 。求终端反射系数、驻波比、电压波节点及波腹点的位置。

解：终端反射系数： $\Gamma_{uL} = \frac{Z_L - Z_c}{Z_L + Z_c} = \frac{100 + j50 - 100}{100 + j50 + 100} = \frac{1}{17}(1 + j4) = \frac{\sqrt{17}}{17}e^{j1.33}$

驻波比： $\rho = \frac{1 + |\Gamma_{uL}|}{1 - |\Gamma_{uL}|} = \frac{1 + \frac{\sqrt{17}}{17}}{1 - \frac{\sqrt{17}}{17}} = \frac{9 + \sqrt{17}}{8} = 1.64$

电压波节点： $d_{\min} = \frac{\varphi(0)\lambda}{4\pi} + \frac{\lambda}{4} + n\frac{\lambda}{2} = (0.36 + 0.5n)\text{m} \quad n=0, \pm 1, \pm 2, \dots$

电压波腹点： $d_{\max} = \frac{\varphi(0)\lambda}{4\pi} + n\frac{\lambda}{2} = (0.11 + 0.5n)\text{m} \quad n=0, \pm 1, \pm 2, \dots$

3. 特性阻抗为 100Ω 的传输线，终端负载为 Z_L ，测得距终端负载 10cm 处是电压波腹点， 20cm 处是相邻的电压波节点，电压驻波比为 3，求终端负载。设入射电压波为 $V^+ = 10e^{-j\beta z}$ ，负载处 $z=0$ ，写出总电压、电流波。

解：由题意可得：

$$d_{\min 1} = \frac{\varphi(0)\lambda}{4\pi} + \frac{\lambda}{4} = 20$$

$$d_{\max 1} = \frac{\varphi(0)\lambda}{4\pi} = 10$$

$$\text{解得 } \lambda = 40\text{cm}, \quad \varphi(0) = \pi$$

$$\text{且 } k = \frac{2\pi}{\lambda} = 5\pi(\text{rad/m})$$

因为驻波比 ρ 为 3

$$\text{所以 } |\Gamma_u| = \frac{\rho-1}{\rho+1} = 0.5$$

$$\Gamma_{uL} = 0.5e^{j\pi} = -0.5$$

$$\text{所以 } \Gamma_u(z) = -0.5e^{j2kz} = -0.5e^{j10\pi z}$$

$$\text{终端负载: } Z_L = Z_C \frac{1+\Gamma_{uL}}{1-\Gamma_{uL}} = \frac{100}{3} = 33.3\Omega$$

$$\beta = k = \frac{2\pi}{\lambda} = 5\pi(\text{rad/m})$$

$$\text{总电压: } U(z) = (1+\Gamma_u(z))V^+ = (1-0.5e^{j10\pi z})10e^{-j5\pi z} = (10e^{-j5\pi z} - 5e^{j5\pi z})\text{V}$$

$$\text{总电流: } I(z) = (1-\Gamma_u(z))\frac{V^+}{Z_C} = (1+0.5e^{j10\pi z})0.1e^{-j5\pi z} = (0.1e^{-j5\pi z} + 0.05e^{j5\pi z})\text{A}$$

43. 传输线终端负载归一化阻抗 $Z_L=0.8+j1.0$ ，计算：

- (1) 驻波系数；
- (2) 离开负载第一个驻波最小点的位置 $d_{\min}$ ；
- (3) 负载反射功率与入射功率之比；
- (4) 作出 $V(z) \sim z/\lambda$ 关系曲线。

$$\text{解: (1) 反射系数: } \Gamma_L = \frac{Z_L - Z_C}{Z_L + Z_C} = \frac{0.8 + j1.0 - 1}{0.8 + j1.0 + 1} = \frac{0.64 + j2}{4.24} = 0.5e^{j1.26}$$

$$\text{驻波系数: } \rho = \frac{1+|\Gamma_L|}{1-|\Gamma_L|} = \frac{1+0.5}{1-0.5} = 3$$

$$(2) \quad d_{\min 1} = \frac{\varphi(0)\lambda}{4\pi} + \frac{\lambda}{4} = \frac{1.26\lambda}{4\pi} + \frac{\lambda}{4} = 0.35\lambda$$

$$d_{\max 1} = \frac{\varphi(0)\lambda}{4\pi} = 0.1\lambda$$

$$(3) \quad \frac{P^r}{P^i} = |\Gamma_L|^2 = 0.25$$

$$(4) \quad V_{\max} = 1 + |\Gamma_L| = 1.5$$

$$V_{\min} = 1 - |\Gamma_L| = 0.5$$

如图所示：

